Algebraic Curves

and their applications in coding theory

Martin Sig Nørbjerg

P6 Project, Group 4.105b, Mathematics

Dept. of Mathematical Sciences
Skjernvej 4A
DK-9220 Aalborg Ø
http://math.aau.dk

Title
Algebraic Curves

Theme
Algebraic Geometry Codes

Project Period
Spring Semester 2022

Project Group
Group 4.105b

Participants
Martin Sig Nørbjerg

Supervisor
Matteo Bonini

Page Numbers
14

Date of Completion
March 5, 2023

Abstract

□

The content of this project is freely available, but publication (with reference) may only be pursued due to agreement with the authors.

Preface

This is a Bachelor project written at the Department of Mathematical Sciences at Aalborg University. As such, it's expected that the reader has taken courses in abstract algebra and linear algebra. It is also expected of the reader to be acquainted with the basics of topology.

The work on the project took place from the 1. of Febuary to the **TODO** of May 2023.

Definitions, propositions, theorems, lemmas, corollaries, and remarks are numbered according to each chapter and consecutively. Equations are numbered separately and likewise with figures and tables. Finally the endings of proofs and examples are marked with ■ and □ respectively.

A table of notation and shorthands is given after the list of contents. Please note that some symbols may be used differently in different chapters.

Aalborg University, March 5, 2023

Martin Sig Nørbjerg
mnarbj20@student.aau.dk

Contents

Preface **ii**

Notation and Shorthands **iv**

1 Error Correcting Codes **1**

1.1 Transmission and Nearest Neighbour Decoding 3

1.2 Bounds on Linear Codes 4

1.2.1 Assymptotic Gilbert-Varshamov Bound 5

2 Algebraic Geoemetry **8**

2.1 Hilbert Basis Theorem 10

2.2 Affine varities 10

Appendices **11**

A Fields and Polynomials **12**

A.1 Algebraicly Closed Fields 13

Notation and Shorthands

Abstract Algebra

$\mathbb{F}_q$	The finite field with $q = p^n$ elements.
$\overline{\mathbb{F}}$	A closed field.
$\Bbbk[\underline{X}_n]$	The multivariate polynomial ring over $\overline{\mathbb{F}}$, also written as $\overline{\mathbb{F}}[x_1, \ldots, x_n]$.
$\langle S \rangle$	The ideal generated by S.

Algebraic Geometry

$\mathbb{A}^n$	The n dimensional affine space.
$Z(S)$	The zero set of $S \subseteq \Bbbk[\underline{X}_n]$.
$\mathcal{T}_Z$	The Zariski topology.
$\mathcal{F}_Z$	The closed sets in the Zariski topology (the affine algebraic sets).

Coding theory

$\mathcal{C}$	A linear code.
G	A generator matrix.
H	A parity check matrix.
d	The Hamming metric.
wt	The Hamming weight.

1 Error Correcting Codes

This chapter is based on [6][Chapter 1 and 3] as well as [9][Chapter 1]. We be considering subspaces of the $\mathbb{F}_q$-vector space $\mathbb{F}_q^n$.

Definition 1.1. Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be a k-dimensional linear subspace, then $\mathcal{C}$ is said to be a $[n,k]_q$ **code** and elements in $\mathcal{C}$ is called **codewords**

If $\mathcal{C}$ is a $[n,k]_q$ code, then $\mathbb{F}_q$ is called the *alphabet*, n the *length*, and k the *dimension* of $\mathcal{C}$. What we shall refer to as a $[n,k]_q$ code is normally refereed to as a linear code, as such we will use the terminology interchangeably.

Moving on we define a metric on $\mathbb{F}_q^n$, named after Richard Hamming, one of the founders of coding theory.

Definition 1.2. The **Hamming metric** $\mathrm{d} : \mathbb{F}_q^n \times \mathbb{F}_q^n \to \mathbb{N}$ is defined as

$$\mathrm{d}(x,y) := |\{i \in \{1,\dots,n\} | x_i \neq y_i\}| \,.$$

Further more let $x \in \mathbb{F}_q^n$, then the **weight** of x is defined as $\mathrm{wt}(x) := \mathrm{d}(x,0)$.

Remark 1.3. It's easy to see that for every choice of $x, y \in \mathbb{F}_q^n$ we have $\mathrm{d}(x,y) = \mathrm{wt}(x-y)$.

It is no coincidence that we refereed to d as a metric, we will prove exactly this in the following proposition.

Proposition 1.4. $(\mathbb{F}_q^n, \mathrm{d})$ *is a metric space, that is*

(i) $\mathrm{d}(x,y) \geq 0$ *and* $\mathrm{d}(x,y) = 0$ *if and only if* $x = y$.

(ii) d *is symmetric, meaning* $\mathrm{d}(x,y) = \mathrm{d}(y,x)$.

(iii) d *complies with the triangle inequality, meaning* $\mathrm{d}(x,z) \leq \mathrm{d}(x,y) + \mathrm{d}(y,z)$.

for all $x, y, z \in \mathbb{F}_q^n$

Proof. Assertation (i) is trivial, and (ii) follows as $\neq$ is a symmetric binary operation. Finally (iii) follows since:

$$\begin{aligned}\mathrm{d}(x,y) &\leq |\{i \in \{1,\dots,n\} | x_i \neq z_i \vee y_i \neq z_i\}| \\ &\leq |\{i \in \{1,\dots,n\} | x_i \neq z_i\}| + |\{i \in \{1,\dots,n\} | y_i \neq z_i\}| = \mathrm{d}(x,z) + \mathrm{d}(z,y).\end{aligned}$$

where the first inequality follows as $x_i \neq y_i$ implies that either $x_i \neq z_i$ or $y_i \neq z_i$. ∎

Definition 1.5. Let $\mathcal{C}$ be a $[n,k]_q$ code, then $d = \min\{\mathrm{d}(a,b) \mid a,b \in \mathcal{C}, a \neq b\}$ is called the **minimum distance** of $\mathcal{C}$, and we say that $\mathcal{C}$ is a $[n,k,d]_q$ code.

Remark 1.6. As $\mathcal{C}$ is a linear subspace we have $d = \min\{\mathrm{wt}(x) | x \in \mathcal{C} \setminus \{0\}\}$, since $y - z \in \mathcal{C}\setminus\{0\}$, for all $y, z \in \mathcal{C}$ such that $y \neq z$. For this reason the minimum distance of $\mathcal{C}$ is also refereed to as the *minimum weight* of $\mathcal{C}$.

The minimum distance of the code $\mathcal{C}$ have a massive impact on $\mathcal{C}$'s error correcting properties, as we will see in Section 1.1.

Example 1.7. Fix $n \in \mathbb{N}$. We will be considering the perhaps simplest, yet usefull, code called the *repetition code* of length n, taking $x \in \mathbb{F}_q$ to $(x, \ldots, x) \in \mathbb{F}_q^n$. As

$$\mathcal{C} = \mathrm{span}_{\mathbb{F}_q}\{(1, \ldots, 1)\}$$

and $\mathrm{d}(c, c') = n$ for all $c, c' \in \mathcal{C}$, the repetition code of length n is a $[n, 1, n]_q$ code. □

Not all linear codes $\mathcal{C}$ are this simple. However as $\mathcal{C}$ is a linear subspace of $\mathbb{F}_q^n$, there exists a $\mathbb{F}_q$-basis of $\mathcal{C}$, such that every $c \in \mathcal{C}$ can be written as a $\mathbb{F}_q$-linear combination of the basis vectors, this idea forms the foundation of the following definition.

Definition 1.8. Let $\mathcal{C}$ be a $[n,k]_q$ code, and suppose $x_1, \ldots, x_k \in \mathbb{F}_q^n$ forms a $\mathbb{F}_q$-basis of $\mathcal{C}$, then

$$G = \begin{bmatrix} x_1 & \cdots & x_k \end{bmatrix}^T \in \mathbb{F}_q^{n \times k}$$

is called a **generator matrix** for $\mathcal{C}$, if $G = \begin{bmatrix} I_k & A \end{bmatrix}$ for some $A \in \mathbb{F}_q^{(n-k)\times k}$ we say that G is in **standard form**.

In many circumstances one gives a generator matrix G instead of writing out $\mathcal{C}$ explicitly. As $\mathcal{C} = \mathrm{row}(G)$.

Definition 1.9. Suppose $\mathcal{C}$ is a $[n,k]_q$ code. Then a matrix $H \in \mathbb{F}_q^{(n-k)\times n}$ such that

$$\mathcal{C} = \{x \in \mathbb{F}_q^n \mid Hx = 0\}$$

is called a **parity check matrix** of $\mathcal{C}$.

A parity check matrix H, gives us yet another way to specify as $\mathcal{C} = \mathrm{null}(H)$.

Lemma 1.10. *Let H be a parity check matrix of a $[n,k]_q$ code, then* $\mathrm{rank}(H) = n - k$.

Proof. Let $T : \mathbb{F}_q^n \to F_q^{n-k}$ be defined as $x \mapsto Hx$. Now as $\dim(\ker(T)) = \dim(\mathcal{C}) = k$, by applying the Fundamental Theorem of Linear Maps [1][Theorem 3.22], it follows that $\mathrm{rank}(H) = \dim(\mathrm{image}(T)) = n - k$. ■

Since every $[n,k]_q$ code is a linear subspace of $\mathbb{F}_q^n$, it makes sense to consider code given as the orthogonal complement of $\mathcal{C}$, which we shall now define, in more detail:

Definition 1.11. Let $\mathcal{C}$ be a $[n,k]_q$ code, then we define the **dual code** of C as:

$$C^\perp := \left\{x \in \mathbb{F}_q^n \middle| \langle x, c\rangle = 0 \text{ for all } c \in \mathcal{C}\right\}$$

where $\langle \cdot, \cdot \rangle$ denotes the usual canonical inner product on $\mathbb{F}_q^n$.

Dual codes have some nice properties, which allows us to use them to study their non-dual counterparts, some of these properties will be shown in the following Lemma, the proof are based on the proofs of Lemma 1.8 and Corollary 1.9 in [3].

Lemma 1.12. *Suppose $\mathcal{C}$ is a $[n,k]_q$ code, with generator matrix G and parity check matrix H then:*

(i) *G is a parity check matrix of $\mathcal{C}^\perp$*

(ii) $\dim(\mathcal{C}^\perp) = n - k$.

(iii) *H is a generator matrix of $\mathcal{C}^\perp$.*

(iv) $\mathcal{C} = (\mathcal{C}^\perp)^\perp$.

Proof. Let $g_1, \dots, g_k$ be the rows of G, and let $x \in \mathbb{F}_q^n$ then $Gx = (\langle x, g_1\rangle, \dots, \langle x, g_k\rangle) = 0$ if and only if $x \perp g_i$ for all $i \in \{1, \dots, k\}$. this implies (i) as $g_1, \dots, g_k$ forms a $\mathbb{F}_q$-basis of $\mathcal{C}$. Assertation (ii) follows from (i) as $\mathcal{C}^\perp$ is the null space of the linear map $x \mapsto Gx$ and $\operatorname{rank}(G) = k$, since the rows of G are $\mathbb{F}_q$-linearly independent, so $\dim(\operatorname{null}(G)) = n - k$. Continuing with (iii): Since H is a parity check matrix, $\operatorname{rank}(H) = n - k$, confer Lemma 1.10. This implies that the $n - k$ rows of H are linearly independent. Thus they form a $\mathbb{F}_q$-basis of some $(n-k)$-dimensional linear subspace $V \subseteq \mathbb{F}_q^n$, which is orthogonal to $\mathcal{C}$ as $Hc = 0$ for all $c \in \mathcal{C}$. Now since V and $\mathcal{C}^\perp$ are both orthogonal to $\mathcal{C}$ both with dimension $n - k$, confer assertion (ii), it follows that $V = \mathcal{C}^\perp$. Finally $\mathcal{C} \subseteq (\mathcal{C}^\perp)^\perp$, as $c \in \mathcal{C}$ implies $c \perp c'$ for all $c' \in \mathcal{C}^\perp$ combining this with $\dim(\mathcal{C}) = \dim(\big(\mathcal{C}^\perp)^\perp\big)$ gives (iv). ■

1.1 Transmission and Nearest Neighbour Decoding

We can now explain how one can use the $[n,k,d]_q$ code $\mathcal{C}$ when transmit data through a noisy communication channel. Given some data $x \in \mathbb{F}_q^k$, which we shall refer to as the *message*, we *encode* said message by computing the corresponding codeword $c := x^T G$, please note the linear transformation $T : \mathbb{F}_q^k \to \mathcal{C}$, defined as $x \mapsto x^T G$, is injective since the rows of G forms a basis of $\mathcal{C}$. Since $c \in \mathbb{F}_q^n$, with $n \geq k$, more data has to be transmitted, however we have added redundant information, which will hopefully help correct any errors induced by the noise in the channel. Since $c \in \mathcal{C}$ is subjected to random noise[1] during transmission we receive some $y \in \mathbb{F}_q^n$, not necessarily equal to c. The main question is now, how to get back the original c or at least an estimate of c say $\hat{c}$. Multiple of such estimates exists, we will however only consider the perhaps most intuitive, named *nearest neighbour decoding*, where we pick the estimate:

$$\hat{c}_{\mathrm{d}} = \underset{c \in \mathcal{C}}{\arg\min}\ \mathrm{d}(c, y)$$

note that this estimate may not be unique. Finally we can obtain an estimate $\hat{x}$ of x, by decoding $\hat{c}$. The decoding of $\hat{c}$ is however highly dependent on the specific code, and will not be discussed in this general setting.

We will now look further at nearest neighbour decoding, we start with the following theorem:

[1] There are many different assumptions one can impose on the distribution this noise, this will however not be considered in this report.

Theorem 1.13. *Let $\mathcal{C}$ be a $[n,k,d]_q$ code, $r = \lfloor \frac{d-1}{2} \rfloor$, and $c, c' \in \mathcal{C}$ such that $c \neq c'$ then*

$$\overline{B_r}(c) \cap \overline{B_r}(c') = \emptyset$$

where $\overline{B_r}(x) = \{y \in \mathbb{F}_q^n \big| d(x,y) \leq r\}$.

Proof. Assume for the sake of contradiction that $\overline{B_r}(c) \cap \overline{B_r}(c') \neq \emptyset$, pick $x \in \overline{B_r}(c) \cap \overline{B_r}(c')$. Now since d complies with the triangle inequality confer Proposition 1.4(iii), we have:

$$\mathrm{d}(c,c') \leq \mathrm{d}(c,x) + \mathrm{d}(x,c') \leq 2r$$

but $2r = 2\lfloor \frac{d-1}{2} \rfloor \leq d-1$, this contradicts the fact that d is the minimum distance of $\mathcal{C}$. ■

Remark 1.14. A natural implication of Theorem 1.13 is that if a maximum of $r = \lfloor \frac{d-1}{2} \rfloor$ errors occur during transmission, and we receive $y \in \mathbb{F}_q^k$ then $\mathrm{d}(y,c) \leq r$, where c is the correct c, and thus $\hat{c}_{\mathrm{d}} = c$. Since $\overline{B_r}(c) \cap \overline{B_r}(c') = \emptyset$ for all $c' \in \mathcal{C} \setminus \{c\}$. Thus a $[n,k,d]_q$ code $\mathcal{C}$ can correct $\lfloor \frac{d-1}{2} \rfloor$ errors, and detect $d-1$ errors, as the received vector, will not be a codeword of $\mathcal{C}$, in this case.

The contents of Remark 1.14 indicate that when we are interested in finding a $[n,k]_q$ code $\mathcal{C}$, with the best error correcting capabilities then we need to find one with a high minimum distance. However upper bounds on the minimum distance given n and k do exist, a few of which will be presented in the following section.

1.2 Bounds on Linear Codes

In this section we will introduce a few select bounds for the parameters of codes, namely the Singleton and assymptotic Gilbert-Varshamov bound. We will start by stating and proving the following Lemma, it's proof will be based upon the ideas presented in the proof of Theorem 2 found in [8].

Proposition 1.15. *let $\mathcal{C}$ be a $[n,k,d]_q$ code, with parity check matrix H then any $d-1$ columns of H are $\mathbb{F}_q$-linearly independent.*

Proof. Let $h_1, \ldots, h_n$ be the columns of H, assume for the sake of contradiction that there exists distinct indices $i_1, i_2, \ldots i_{d-1}$ such that $h_{i_1}, h_{i_2}, \ldots, h_{i_{d-1}}$ are $\mathbb{F}_q$-linearly dependent. Then we may pick $c \in \mathcal{C} \setminus \{0\}$ with $\mathrm{wt}(c) \leq d-1$, by setting $c_k = 0$ for all $k \notin \{i_1, i_2 \ldots, i_{d-1}\}$, but as $h_{i_1}, h_{i_2}, \ldots, h_{i_{d-1}}$ are $\mathbb{F}_q$-linearly dependent, we can find at least one $k \in \{i_1, i_2 \ldots, i_{d-1}\}$ such that we can pick $c_k \neq 0$. Thus we have $\mathrm{wt}(c) \leq d-1$, which contradict the assumption that d is the minimum distance of $\mathcal{C}$. ■

The Singleton bound now naturally follows.

Corollary 1.16 (Singleton Bound)**.** *Let $\mathcal{C}$ be a $[n,k,d]_q$ code then $d-1 \leq n-k$.*

Proof. Let H be a parity check matrix of $\mathcal{C}$, then $d-1 \leq \mathrm{rank}(H)$ as any collection of $d-1$ columns of H are $\mathbb{F}_q$-linearly independent, confer Proposition 1.15. The rest follows from the fact that $\mathrm{rank}(H) = n-k$ confer Lemma 1.10. ■

Remark 1.17. Rearranging the Singleton bound to obtain $d + k \leq n + 1$ makes the trade-of between minimum distance and dimension, even more obvious, as such we can either wish to find a code with a high minimum distance d or a high dimension k, relative to it's length n, but not both!

Definition 1.18. If a $[n, k, d]_q$ code $\mathcal{C}$ meets the singleton bound, that is $d - 1 = n - k$, then $\mathcal{C}$ is called a **maximum distance seperable (MDS)** code.

We have actually already seen a MDS code, mainly the repetition code of length n, as seen in example 1.7 The following example is based on elements of [9][Subsection 1.2.1] and [3][chapter 2], and it will be our second example of a MDS code.

Example 1.19. [Reed-Solomon Codes] Let $q = p^m$, where p is prime, and $n, k \in \mathbb{N}$ such that $1 \leq k \leq n \leq q$. As such we may find distinct $P_1, \ldots, P_n \in \mathbb{F}_q$ and define the set $\mathcal{P} := \{P_1, \ldots, P_n\}$. Finally we define the vector space $L_k := \{f \in \mathbb{F}_q[X] \mid \deg(f) \leq k - 1\}$, we have $0 \in L_k$, since we use the convention that $\deg(0) = 0$, confer remark A.6. Then the evaluation map:

$$\mathrm{Ev}_{\mathcal{P}} \colon L_k \to \mathbb{F}_q^n, \quad f \mapsto (f(P_1), \ldots, f(P_n))$$

is injective, since $\mathrm{Ev}_{\mathcal{P}}$ is a linear map and $f \in \mathbb{F}_q[X] \backslash \{0\}$ can have at most $\deg(f)$ roots, so $\mathrm{Ev}_{\mathcal{P}}(f) = 0$ if and only if $f = 0$. We define the Reed-Solomon code of degree k as $\mathcal{C} := \mathrm{Ev}_{\mathcal{P}}(L_k)$, we will show that $\mathcal{C}$ is a $[n, k, n - k + 1]_q$ code. As $1, X, \ldots, X^{k-1}$, forms a basis of L_k, we see that $\mathcal{C}$ has dimension k, further more that

$$G = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ P_1 & P_2 & \cdots & P_n \\ \vdots & \vdots & \ddots & \vdots \\ P_1^{k-1} & P_2^{k-1} & \cdots & P_n^{k-1} \end{bmatrix}$$

is a generator matrix for $\mathcal{C}$. Now every codeword $c \in \mathcal{C}$, has weight w, then f vanishes in $n - w$ of the points and thus we have $n - w \leq \deg(f) \leq k - 1$. In the special case $w = d$ where d is the minimum distance of $\mathcal{C}$, we see that $n - d \leq k - 1$ rearranging we see that $n - k \leq d - 1$, combining this with the Singleton bound (Corollary 1.16) we get that $n - k = d - 1$, and thus $\mathcal{C}$ is a MDS code, with minimum distance $n - k + 1$. This is of cause a nice property to have, however these codes aren't the solution to all of our problems as we require $n \leq q$, so their length is limited. □

We will later show that Reed-Solomon codes are a special case of our main object of study, namely algebraic geometry codes.

1.2.1 Assymptotic Gilbert-Varshamov Bound

The following subsection is based on [6][Section 2.1, 2.8 and 2.10] and [3][Chapter 5]. We start by stating and proving a few technical results as well as the non asymptotic Gilbert bound, and introduce some new notation, along the way, before reaching the main result of this subsection, namely the Gilbert-Varshamov bound.

Lemma 1.20. *Let $x \in \mathbb{F}_q^n$, then $\left|\overline{B_r}(x)\right| = V_q(n, r)$ where $V_q(n, r) = \sum_{i=0}^{r} \binom{n}{i} (q - 1)^i$.*

Proof. Let $S_i(x) = \{x' \in \mathbb{F}_q^n \big| \mathrm{d}(x, x') = i\}$ for $i = 0, 1, \ldots, r$. Then $S_0(x), S_1(x), \ldots, S_r(x)$ are all pairwise disjoint and $\overline{B_r}(x) = \bigcup_{i=0}^{r} S_i(x)$. This gives

$$\left|\overline{B_r}(x)\right| = \sum_{i=0}^{r} |S_i(x)| \tag{1.1}$$

Now $|S_i(x)| = \binom{n}{i}(q-1)^i$, since $x' \in S_i$ differ from x in i out of n entries. The result follows immediately by combining this with Equation (1.1). ■

Moving forward we will use the notation $B_q(n, d)$ be the maximum number of codewords in a linear code, of length n and minimum distance at least d.

Proposition 1.21. *If $\mathcal{C}$ is linear code over $\mathbb{F}_q$ of length n, with minimum distance d and $B_q(n, d)$ codewords then:*

$$\mathbb{F}_q^n = \bigcup_{c \in \mathcal{C}} \overline{B_{d-1}}(c)$$

Proof. det her bevis skal du se bort fra. Assume for the sake of contradiction that there exists at least one $y \in \mathbb{F}_q^n$, such that $y \notin \bigcup_{c \in \mathcal{C}} \overline{B_{d-1}}(c)$. Now pick

$$x = \underset{y \in \mathbb{F}_q^n \setminus \bigcup_{c \in \mathcal{C}} \overline{B_{d-1}(c)}}{\arg\min} (d(y, \mathcal{C}))$$

then $kx \notin \bigcup_{c \in \mathcal{C}} \overline{B_{d-1}}(c)$ for all $k \in \mathbb{F}_q$ **TODO**. This in turn imply that $\mathcal{C}' = \mathcal{C} \cup \operatorname{span}\{x\}$ is another linear code, with minimun distance d and length n, however $|\mathcal{C}'| > |\mathcal{C}|$, which contradicts the fact that $\mathcal{C}$ had the maximum number of codewords. ■

We are now able to state and prove the Gilbert Bound which we will use to prove the asymptotic Gilbert-Varshamov bound.

Corollary 1.22 (Gilbert Bound)**.** *Suppose $V_q(n, r)$ is defined as in Lemma 1.20 then:*

$$B_q(n, d) \geq \frac{q^n}{V_q(n, d-1)}$$

Proof. Let $\mathcal{C}$ be a linear code of length n, with minimum distance d and $B_q(n, d)$ codewords, then $\mathbb{F}_q^n = \bigcup_{c \in \mathcal{C}} \overline{B_{d-1}}(c)$ by Proposition 1.21, combining this with Lemma 1.20, we get

$$q^n = \left|\mathbb{F}_q^n\right| \leq \sum_{c \in \mathcal{C}} \left|\overline{B_q}(c)\right| = B_q(n, d) V_q(n, d-1). \qquad ■$$

Continuing we define the following function which will used in the asymptotic Gilbert-Varshamov bound.

Definition 1.23. The function $H_q : [0, (q-1)/q] \to \mathbb{R}$ defined as $H_q(0) = 0$ and $H_q(x) = x \log_q(q-1) - x \log_q(x) - (1-x) \log_q(1-x)$ when $x \neq 0$ is called the **q-ary entropy function**

We will now state Lemma 2.10.3, from [6], but we will omit it's computation heavy proof.

Lemma 1.24. *Let $0 \leq \delta \leq (q-1)/q$ where $q \geq 2$, then $\lim_{n \to \infty} \frac{1}{n} \log_q(V_q(n, \lfloor \delta n \rfloor)) = H_q(\delta)$.*

Moving on we introduce two invariants which can be used to gauge the quality of a code.

Definition 1.25. Let $\mathcal{C}$ be a $[n,k,d]_q$ code, then the invariants $R(\mathcal{C}) := k/n$ and $\delta(\mathcal{C}) := d/n$ is called the **transmission rate** and **relative distance** of $\mathcal{C}$ respectively.

Following the definition a natural question arise, namely why would one consider the invariants $R(\mathcal{C})$ and $\delta(\mathcal{C})$ instead of using n, k and d directly, when gauging the quality of a $[n,k,d]_q$ code? To answer this consider repetition code of length n, as in example 1.7, when n increases then so will d since $d = n$, however $\delta(\mathcal{C})$ will stay constant as n increases. Also as we are still only able to encode a single symbol we are sacrificing transmission rate.

As $B_q(n,d)$ is the maximum number of codewords in some linear code $\mathcal{C}$, with length n and minimum distance at least d, which is a subspace of $\mathbb{F}_q^n$, we have $\dim(\mathcal{C}) = \log_q(B_q(n,d))$. Thus it makes sense to define what we shall reffer to as the *maximal rate*:

$$R^*(n,d) = \frac{\log_q(B_q(n,d))}{n}.$$

We will now investigate the behavior of $R^*(n,d)$ as $n \to \infty$, for this we define, the function $\alpha_q^{lin} : [0,1] \to [0,1]$ as a function of a relative distance δ, more specifically:

$$\alpha_q^{lin}(\delta) = \limsup_{n\to\infty} R^*(n, \delta n),$$

Finally we can state and prove the Assymptotic Gilbert-Varshamov[2] Bound.

Theorem 1.26 (Assymptotic Gilbert-Varshamov Bound)**.** *Let $0 \le \delta \le (q-1)/q$, then*

$$\alpha_q^{lin}(\delta) \ge 1 - H_q(\delta).$$

Proof. First note that $B_q(n, \delta n) = B_q(n, \lceil \delta n \rceil)$, as $B_q(n,d)$ is the maximum number of codewords of a code of length n with minimum distance at least d. Thus we have

$$\begin{aligned}\alpha_q^{lin}(\delta) = \limsup_{n\to\infty} R^*(\delta) &= \limsup_{n\to\infty} \frac{\log_q(B_q(n, \lceil \delta n \rceil))}{n} \\ &\ge \limsup_{n\to\infty} \frac{\log_q\left(\frac{q^n}{V_q(n, \lceil \delta n \rceil - 1)}\right)}{n} \ge \limsup_{n\to\infty} 1 - \frac{\log_q(V_q(n, \lfloor \delta n \rfloor)}{n}\end{aligned}$$

where the first inequality follows from the Gilbert bound (Corollary 1.22) and the last inequality follows since $\lceil \delta n \rceil - 1 \le \lfloor \delta n \rfloor$ implies $\frac{q^n}{V_q(n, \lceil \delta n \rceil - 1)} \ge \frac{q^n}{V_q(n, \lfloor \delta n \rfloor)}$. The rest follows by applying Lemma 1.24, as

$$\lim_{n\to\infty} \frac{\log_q(V_q(n, \lfloor \delta n \rfloor)}{n} = H_q(\delta) \implies \limsup_{n\to\infty} \frac{\log_q(V_q(n, \lfloor \delta n \rfloor)}{n} = H_q(\delta). \qquad \blacksquare$$

[2]The Varshamov bound is another bound for the parameters of linear codes, which can be used to prove the theorem as well.

2 Algebraic Geoemetry

Algebraic geometry is the study of zero sets of polynomials, over an arbitrary but closed field $\Bbbk$, please confer definition A.7. The ring of multivariate polynomials with coefficients in $\Bbbk$ is denoted as $\Bbbk[X_1, \ldots, X_n]$ or $\Bbbk[\underline{X}_n]$. First we introduce some basic definitions and ground work, before moving onto some more interesting results.

Definition 2.1. The n-dimensional **affine space** over $\Bbbk$ is defined as $\mathbb{A}^n := \{(a_1, \ldots, a_n) \mid a_i \in \Bbbk\}$.

Suppose $P = (a_1, \ldots, a_n) \in \mathbb{A}^n$ and $f \in \Bbbk[\underline{X}_n]$, then P is called a *point* and we may write $f(P)$ instead of $f(a_1, \ldots, a_n)$.

Definition 2.2. Let $S \subseteq \Bbbk[\underline{X}_n]$, then we define the **zero set** of S as

$$Z(S) = \{P \in \mathbb{A}^n | f(P) = 0 \text{ for all } f \in S\}.$$

and $Z(S)$ is said to be an **affine algebraic set** in $\mathbb{A}^n$.

Remark 2.3. It's easy to see that if $S \subseteq \Bbbk[\underline{X}_n]$ and $T \subseteq S$, then $Z(S) \subseteq Z(T)$, since $P \in Z(S)$ implies $f(P) = 0$ for all $f \in S$ and in particular for all $f \in T$.

When $S = \{f_1, \ldots, f_m\} \subseteq \Bbbk[\underline{X}_n]$, we may write $Z(S)$ as $Z(f_1, \ldots, f_m)$. If $f \in \Bbbk[\underline{X}_n] \backslash \Bbbk$ we call $Z(f)$ a *hypersurface*. We will now show that every affine algebraic set, is the zero set of some ideal in $\Bbbk[\underline{X}_n]$

Lemma 2.4. *Let $S \subseteq \Bbbk[\underline{X}_n]$, then $Z(S) = Z(\langle S \rangle)$.*

Proof. Since $S \subseteq \langle S \rangle$, it follows from remark 2.3 that $Z(\langle S \rangle) \subseteq Z(S)$. Now suppose $p \in Z(S)$, then $f(p) = 0$ for all $f \in S$, now let $g \in \langle S \rangle$, thus g can be written as

$$g = \sum_{i=1}^{n} h_i f_i$$

for some $n \in \mathbb{N}, h_i \in \mathbb{F}[\underline{x}_n], f_i \in S$, now evaluating g at the point p gives:

$$g(p) = \sum_{i=1}^{n} h_i(p) f_i(p) = 0$$

since $f_i(p) = 0$ for all $i = 1, \ldots, n$. ■

We will now define a topology on $\mathbb{A}^n$, where the affine algebraic sets form the closed sets.

Definition 2.5. The set $\mathcal{T}_Z = \{Z(S)^c \mid S \subseteq \Bbbk[\underline{X}_n]\}$ is called the **Zariski topology**. We will denote the set of closed sets, in the topology $\mathcal{T}_Z$, with $\mathcal{F}_Z = \{Z(S) \mid S \subseteq \Bbbk[\underline{X}_n]\}$.

We shall now prove that the Zariski topology is indeed actually a topology over $\mathbb{A}^n$. We do however first require the following proposition.

Proposition 2.6. *Let $S, T \subseteq \Bbbk[\underline{X}_n]$ and $\{S_i\}_{i\in\mathcal{I}}$ be a family of sets of polynomials in $\Bbbk[\underline{X}_n]$ then:*

(i) $\mathbb{A}^n \in \mathcal{F}_Z$ *and* $\emptyset \in \mathcal{F}_Z$

(ii) $Z(S) \cup Z(T) \in \mathcal{F}_Z$.

(iii) $\bigcap_{i\in\mathcal{I}} Z(S_i) \in \mathcal{F}_Z$.

Proof. We will prove each claim individually.

(i) We have $\mathbb{A}^n = Z(\emptyset)$ and $\emptyset = Z(\Bbbk^*)$.

(ii) It is the case that $Z(S) \cup Z(T) = Z(S \cdot T)$, where $S \cdot T = \{f \cdot g \mid f \in S, g \in T\}$. If $p \in Z(S) \cap Z(T)$, then $h(p) = f(p) \cdot g(p) = 0$ for all $f \in S$ and $g \in T$. Thus $Z(S) \cap Z(T) \subseteq Z(S \cdot T)$. On the otherhand if $p \in Z(S \cdot T)$, then $h(p) = f(p) \cdot g(p) = 0$ for all $f \in S, g \in T$, assuming $p \notin Z(S)$, then there exists $f \in S$ such that $f(p) \neq 0$ however this implies that $g(p) = 0$ for all $g \in T$, since $\Bbbk$ is a domain, and it therefore follows $Z(S \cdot T) \subseteq Z(S) \cap Z(T)$.

(iii) Finally we have $\bigcap_{i\in\mathcal{I}} Z(S_i) = Z\left(\bigcup_{i\in\mathcal{I}} S_i\right)$, which follows directly from definition 2.2. ■

Corollary 2.7. $(\mathbb{A}^n, \mathcal{T}_Z)$ *is a topological space.*

Proof. Let $X, Y \in \mathcal{T}_Z$, then there exists $S, T \subseteq \Bbbk[\underline{X}_n]$ such that $X = \mathbb{A}^n \backslash Z(S), Y = \mathbb{A}^n \backslash Z(T)$, thus it follows that

$$X \cap Y = \mathbb{A}^n \backslash (Z(S) \cap Z(T)) = \mathbb{A}^n \backslash Z(S \cdot T) = Z(S \cdot T)^c \in \mathcal{T}_Z.$$

confer proposition 2.6. Let $\{X_i\}_{i\in\mathcal{I}}$ be a family of sets in $\mathcal{T}_Z$, then there exists a family $\{S_i\}_{i\in\mathcal{I}}$, such that $X_i = \mathbb{A}^n \backslash Z(S_i)$ for all $i \in \mathcal{I}$. It therefore follows that

$$\bigcup_{i\in\mathcal{I}} X_i = \bigcup_{i\in\mathcal{I}} \mathbb{A}^n \backslash Z(S_i) = \mathbb{A}^n \backslash \bigcap_{i\in\mathcal{I}} Z(S_i) = \mathbb{A}^n \backslash Z\left(\bigcup_{i\in\mathcal{I}} S_i\right) \in \mathcal{T}_Z$$

which follows from De Morgan Law for set differences and proposition 2.6. Finally $\emptyset, \mathbb{A}^n \in \mathcal{F}_Z$, implies that $\emptyset^c = \mathbb{A}^n \in \mathcal{T}_Z$ as well as $(\mathbb{A}^n)^c = \emptyset \in \mathcal{T}_Z$. ■

Example 2.8. TODO □

Definition 2.9. Let $X \subseteq \mathbb{A}^n$, then the **ideal of** X is defined as

$$I(X) := \{f \in \Bbbk[\underline{X}_n] \mid f(p) = 0 \text{ for all } p \in X\}$$

2.1 Hilbert Basis Theorem

Definition 2.10. Let R be a ring, and supposse all ideals of R are finitely generated, then R is said to be **Noetherian**.

Remark 2.11. Every field $\mathbb{F}$ is Noetherian, since $\emptyset = \langle \emptyset \rangle$ and $\mathbb{F} = \langle 1 \rangle$ are the only ideals. Another even more obvious example are principal ideal domains.

We will now state and prove the main result of this section,

Theorem 2.12 (Hilbert basis theorem)**.** *Let R be Noetherian, then $R[\underline{x}_n]$ is Noetherian.*

Proof. **TODO** ■

Corollary 2.13. *Let $\mathbb{F}$ be an arbitrary field, then $\mathbb{F}[\underline{x}_n]$ is Noetherian.*

Proof. Follows directly from remark 2.11 and theorem 2.12. ■

Theorem 2.14. *Suppose $S \subseteq \Bbbk[\underline{X}_n]$, then there exists $n \in \mathbb{N}$ and $f_1, \ldots, f_n \in \Bbbk[\underline{X}_n] \backslash \Bbbk$ such that*

$$Z(S) = \bigcap_{i=1}^{n} Z(f_i)$$

Proof. The ring $\Bbbk[\underline{X}_n]$ is noetherian, confer corollary 2.13, and since $Z(S) = Z(\langle S \rangle)$ confer lemma 2.4, it follows that there exists $f_1, \ldots, f_n \in \Bbbk[\underline{X}_n]$, such that

$$Z(S) = Z(\langle f_1, \ldots, f_n \rangle) = Z(f_1, \ldots, f_n) = \bigcap_{i=1}^{n} Z(f_i). \qquad ■$$

2.2 Affine varities

Definition 2.15 (Affine and quasi-affine varieties)**.** Let $X \in \mathcal{F}_Z$, X is called an affine variety if there exists no

Appendices

A Fields and Polynomials

In this appendix we start by defining multivariate polynomials over an arbetary ring R, before moving on to algebraically closed fields and algebraic closures. The following definitions will be based on those found in [2][Section 1.1] and [4][Lecture 1]

Definition A.1. Let R be a ring, then we define the **ring of multivariate polynomials** with $n \geq 1$ variables over R inductively as $R[X_1, \ldots, X_n] := R[X_1, \ldots, X_{n-1}][X_n]$ with $R[X_1]$, defined the usual way.

Remark A.2. We will often use the notation $R[\underline{X}_n]$, instead of $R[X_1, \ldots, X_n]$.

Alternatively if $n = 1, 2$ or 3 we may write $R[X]$, $R[X, Y]$ or $R[X, Y, Z]$ instead of $R[\underline{X}_n]$. We will now introduce some special types of multivariate polynomials, with the end goal of being able to define the degree of a arbitrary multivariate polynomial.

Definition A.3. Multivariate polynomials of the form $g = \prod_{i=1}^{n} X_i^{k_i}$ where $k_i \in \mathbb{N}$ are called **monomials**. The degree of the monomial g is defined as $\deg(g) := \sum_{i=1}^{n} k_i$.

We will now consider polynomials given as a linear combinations of monomials of the same degree.

Definition A.4. Let $h \in R[\underline{X}_n]$. Suppose that there exists $d, k \in \mathbb{N}$, as well as $g_1, \ldots g_n \in R[\underline{X}_n]$ all monomials of degree d, and $a_1, \ldots, a_n \in R$ such that $h = \sum_{i=1}^{n} a_i g_i$. Then h is called **homogenous** and we define $\deg(h) := d$.

Finally we are able to define the degree of an arbitrary multivariate polynomial.

Definition A.5. Let $f \in R[\underline{X}_n]$ then there exists some $d \in \mathbb{N}$ and $h_1, \ldots, h_d \in R[\underline{X}_n]$, all homogenous, with $\deg(h_i) = i$, such that $f = \sum_{i=1}^{d} h_i$. Then we define the **degree** of f to be $\deg(f) = d$.

Remark A.6. Since the $0 \in R[\underline{X}_n]$, can be written as the empty sum of homogenous polynomials, we will be using the convention that $\deg(0) = 0$.

It is worth noting that if f is a monomial or a homogenous polynomial, the degree defined in definition A.5 agrees with the degree in A.3 or A.4 respectively. Finally if $n = 1$, the degree of f defined in definition A.5 also coresponds to the standard definition, since X^k where $k \in \mathbb{N}$ are the only monomials in $R[X]$.

A.1 Algebraicly Closed Fields

Unless otherwise specified the definitions and results in this section will be based on those presented in [5][Sections 3.1 to 3.3]. Most of the theory of algebraic geometry assumes that we are working over fields, where polynomials always has at least one root. It is thus natural to introduce the following definition.

Definition A.7. Let $\mathbb{F}$ be a field, if there for all $f \in \mathbb{F}[X] \backslash \mathbb{F}^*$, exists an $\alpha \in \mathbb{F}$ such that $f(\alpha) = 0$. Then $\mathbb{F}$ is said to be **algebraically closed**.

Remark A.8. We often denote a arbitrary closed field with the letter $\Bbbk$.

When applying algebraic geometry to error correcting codes, we will be working with finite fields, these are however not algebraically closed, as we will show in the following proposition.

Proposition A.9. *Let $\mathbb{F}_q$ be a finite field, then $\mathbb{F}_q$ is not algebraically closed*

Proof. Enumerate the distinct elements in $\mathbb{F}_q$ as $a_1, \ldots, a_q$. Consider the polynomial $f = 1 + \prod_{k=1}^{q}(a_k - x) \in \mathbb{F}_q[X] \backslash \mathbb{F}_q^*$. If $\alpha \in \mathbb{F}_q$ then

$$f(\alpha) = 1 + \prod_{k=1}^{q}(a_k - \alpha) = 1,$$

since $\alpha = a_i$ for some $i \in \{1, \ldots, q\}$, thus f has no roots in $\mathbb{F}$. ■

Definition A.10. Let $\mathbb{K}$ be a field, if $\mathbb{F} \subseteq \mathbb{K}$ is a subfield, we say that $\mathbb{K}$ is a **field extension** of $\mathbb{F}$, also written $\mathbb{K}/\mathbb{F}$. The field extension $\mathbb{K}/\mathbb{F}$ is called an **algebraic closure** of $\mathbb{F}$, if $\mathbb{K}$ is algebraically closed.

Remark A.11. This definition, of algebraic closures, is less strict than the definition given in [5], where they require $\mathbb{K}$ to be the smallest algebraically closed field extension, however our defintion is sufficient for the scope of this project.

In the previous proposition we showed that the arbitrary finite field $\mathbb{F}_q$ is not algebraically closed. This is a problem if we want to apply the theory of algebraic geometry, to error correcting codes which operate over finite fields, since we always assume our ground field to be algebraically closed. However if an algebraic closure of $\mathbb{F}_q$ exists, one can instead work with this as the ground field, minimizing the extend of the problem. In the following theorem we will show that such an algebraic closure of a finite field $\mathbb{F}_p$, where p is a prime, exists. The proof will be based on the ideas found in [10][Proof of Theorem 2.2.6], as well as [7][Remark 4.6.8].

Theorem A.12. *Let p be prime, then $\overline{\mathbb{F}}_p = \bigcup_{n=1}^{\infty} \mathbb{F}_{p^n}$ is an algebraic closure of $\mathbb{F}_p$.*

Proof. Let $f \in \overline{\mathbb{F}}_p[X]$ then there exists $n \in \mathbb{N}$ such that $f \in \mathbb{F}_{p^n}[X]$. We can without loss of generality assume f to be irreducible[1]. Then $F := \mathbb{F}_p[X]/\langle f \rangle$ is a finite field, with $p^{\deg(f)}$ elements and thus $F \cong \mathbb{F}_{p^{\deg(f)}}$, further more we have $[X] \in F$ and $f([X]) = 0 \in F$, this combined with the fact that F is isomorphic to a subfield of $\overline{\mathbb{F}}_p$ implies that f must have a root in $\overline{\mathbb{F}}_p$. ■

[1] Otherwise write f as a product of irreducible polynomials, and continue the proof, by replacing f with one of its irreducible factors.

Bibliography

[1] Axler, S. (2015). *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Springer International Publishing, 3rd edition.

[2] Fulton, W. (2008). Algebraic curves: An introduction to algebraic geometry. avalible at `https://dept.math.lsa.umich.edu/~wfulton/CurveBook.pdf`.

[3] Giulietti, M. (2003). Notes on algebraic-geometric codes. Lecture notes from a series of lectures, given in May 2003.

[4] Göttsche, L. (2012). Algebraic geometry lectures. Video recordings of algebraic geometry lectures, given at ICTP Mathematics, can be found at `https://www.youtube.com/playlist?list=PLLq_gUfXAnkkeQKfjfWwyJP8eUV08Kygd`.

[5] Göttsche, L. (2016). Introduction to algebra. Lecture notes, can be found at: `https://users.ictp.it/~gottsche/`.

[6] Huffman, W. C. and Pless, V. (2003). *Fundementals of Error-Correcting Codes*. Cambrige University Press, 1st edition.

[7] Lauritzen, N. (2003). *Concrete Abstract Algebra: From Numbers to Gröbner Bases*. Cambridge University Press, 1st edition.

[8] Shum, K. and Zhang, B. (2016). The singleton bound and reed-solomon code. avalible at `https://piazza.com/class_profile/get_resource/isgy6spmwwm3ba/itzo5as3bbw7kk`.

[9] Tsfasman, M., Vlăduţ, S., and Nogin, D. (2007). *Algebraic Geometry Codes: Basic Notations*. American Mathematical Society, 1st edition.

[10] Weintraub, S. H. (2020). *Galois Theory*. Springer, 2nd edition.